WILBERT ANDREW YONATHAN

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wilbert-andrew-yonathan.vercel.app | huggingface.co/Wizzas | Sepang, Selangor
Available for an internship from 29 March 2027 for up to five months

EDUCATION

Xiamen University Malaysia Sep 2023 – Sep 2027 (Expected)
Bachelor of Engineering in Artificial Intelligence | CGPA: 3.77/4.00

SKILLS

Programming Languages: Python, C, C++, JavaScript, SQL
ML & Computer Vision: TensorFlow/Keras, PyTorch, scikit-learn, Ultralytics, OpenCV, YOLOv8, DINOv2
Generative AI & NLP: LLMs, RAG, LangChain, LangGraph, Hugging Face Transformers, Qwen, BGE Embeddings
Data & Databases: NumPy, pandas, SQLite, Supabase (PostgreSQL, Auth, RLS)
Web & APIs: FastAPI, React (React.js), HTML/CSS, Gradio, REST APIs
Cloud & Deployment: AWS Lightsail, AWS Budgets, Docker, Docker Compose, Linux (Ubuntu), GitHub Actions (CI/CD), Hugging Face Spaces (ZeroGPU), Render, Vercel, Caddy
Tools & Automation: Git, pytest, n8n, Webhooks, Resend
Languages: English (professional working), Mandarin and Malay (conversational), Indonesian and Hokkien (native)

PROJECT EXPERIENCE

PaddyLiteX: Paddy Disease Detection (Final-Year Thesis) | GitHub | Live demo Apr 2026 – Present
- Developed a DenseLiteX-SPPF-C2PSA YOLOv8n detector for nine paddy classes, obtaining 81.67% mAP@0.5 on 803 test images, 0.88 percentage points above the original YOLOv8n baseline with 4.24% fewer parameters.
- Reduced parameters by 68.01% and GFLOPs by 72.59% compared to the initial YOLOv8n-DenseNet121 method; conducted ablation studies on different variants and deployed demos on Render and Hugging Face Spaces.

DefectRAG: Visual Anomaly Inspection and RAG | GitHub | Live demo Aug 2026 – Nov 2026
- Built a computer vision anomaly detection system in Python and PyTorch that finds defective products from normal photos only; fine-tuned DINOv2 to reach 0.971 AUROC on VisA (from 0.950) and 0.992 on MVTec AD.
- Built a retrieval-augmented generation (RAG) assistant with LangChain and LangGraph; fine-tuned an LLM (Qwen2.5-3B, QLoRA, 8 GB GPU), raising correct sources from 42% to 89% on 100 held-out questions.
- Deployed a live app on Hugging Face: React and FastAPI, Supabase authentication (TOTP MFA, row-level security), 490+ automated tests, GitHub Actions CI/CD, and n8n email automation on AWS Lightsail with Docker.

MarineLite: Marine Organism Detection | GitHub | Live demo Jun 2026 – Jul 2026
- Integrated GhostConv and SimSPPF into DU-MobileYOLO, improving mAP@0.5 from 74.23% to 75.36% on URPC2020 dataset while reducing parameters from 4.70M to 4.37M.
- Compared five ablation variants and deployed public demos on Hugging Face and Render.

dPickleball: Reinforcement Learning Agent | GitHub | Competition video May 2026 – Jul 2026
- Trained an independent Proximal Policy Optimization (PPO) agent in Unity ML-Agents, testing reward and penalty configurations within the team's curriculum-learning and league self-play pipeline.
- Contributed a candidate policy to round-robin evaluation; the team placed 2nd of 13 in the class competition.

WORK EXPERIENCE

Student Registration & Operations Assistant, Xiamen University Malaysia Apr 2025
- Assisted registration logistics and directed students within a team serving 800+ foundation students.
- Organized university assets and materials for event operations and storage.

LEADERSHIP AND ACTIVITIES

Head of Photography & Videography, Love in Action Feb 2026 – Mar 2026
Led the media team for a charity project serving 150+ participants; scripted, directed and edited the final recap video.

Head of General Affairs, XMUM AI Club Apr 2025 – Oct 2025
Handled budget, procurement and corporate-partner merchandise for a national AI and digital sports tournament.

Assistant, General Affairs, AIESEC XMUM Sep 2023 – Sep 2025
Managed room bookings, venue setups and organized equipment used for weekly meetings and club events.

AWARDS

- Dean's List, three consecutive semesters
- XMUM Merit Scholarship, four-year undergraduate program, 2023–2027

REFERENCES

Shaidah Binti Jusoh, Assistant Professor, Thesis Supervisor | shaidah.jusoh@xmu.edu.my | 019-6544207
Goh Sim Kuan, Assistant Professor, Internship Coordinator | simkuan.goh@xmu.edu.my | 012-7660576